

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. CURRENT SYSTEM BASELINE (MVP):

The current Minimum Viable Product (MVP) operates efficiently as a local web application generating 5-panel short comics. It proves the concept that chained Generative AI models can automate visual publishing. However, the architecture is designed to be highly extensible.

3. FUTURE SCALABILITY INITIATIVES :

To evolve ComicCraft from an academic project into a commercial-grade SaaS (Software as a Service) platform, the following scaling strategies are proposed:

- **Cloud Deployment & Serverless Architecture:** Transitioning the FastAPI backend from local Uvicorn hosting to managed cloud platforms like AWS Lambda or Google Cloud Run. This will allow the application to handle thousands of concurrent user requests effortlessly.
- **Database & User Authentication:** Integrating Firebase or PostgreSQL to introduce user accounts. This allows users to maintain a persistent cloud library of their generated comic books instead of relying solely on local downloads.
- **Extended Narrative Generation:** Upgrading the system logic to support recursive AI prompting. This would enable the generation of 20+ page graphic novels with complex, branching chapters rather than being limited to short strips.
- **Advanced AI Consistency (LoRA):** Incorporating Low-Rank Adaptation (LoRA) models within the Stable Diffusion pipeline. This major future upgrade will ensure that the protagonist's facial features and clothing remain 100% strictly consistent across every single panel, a current limitation in standard text-to-image AI.

- **Interactive Editing Suite:** Developing a frontend canvas that allows users to manually edit the AI-generated text or swap out specific images before initiating the final PDF download.